CPSC 368: Connecting to Oracle

Connecting to the Oracle Server

As the department's Oracle database is a protected host, we must first tunnel into the department before accessing the database. The tunnel must remain open throughout the duration you want to connect to the database.

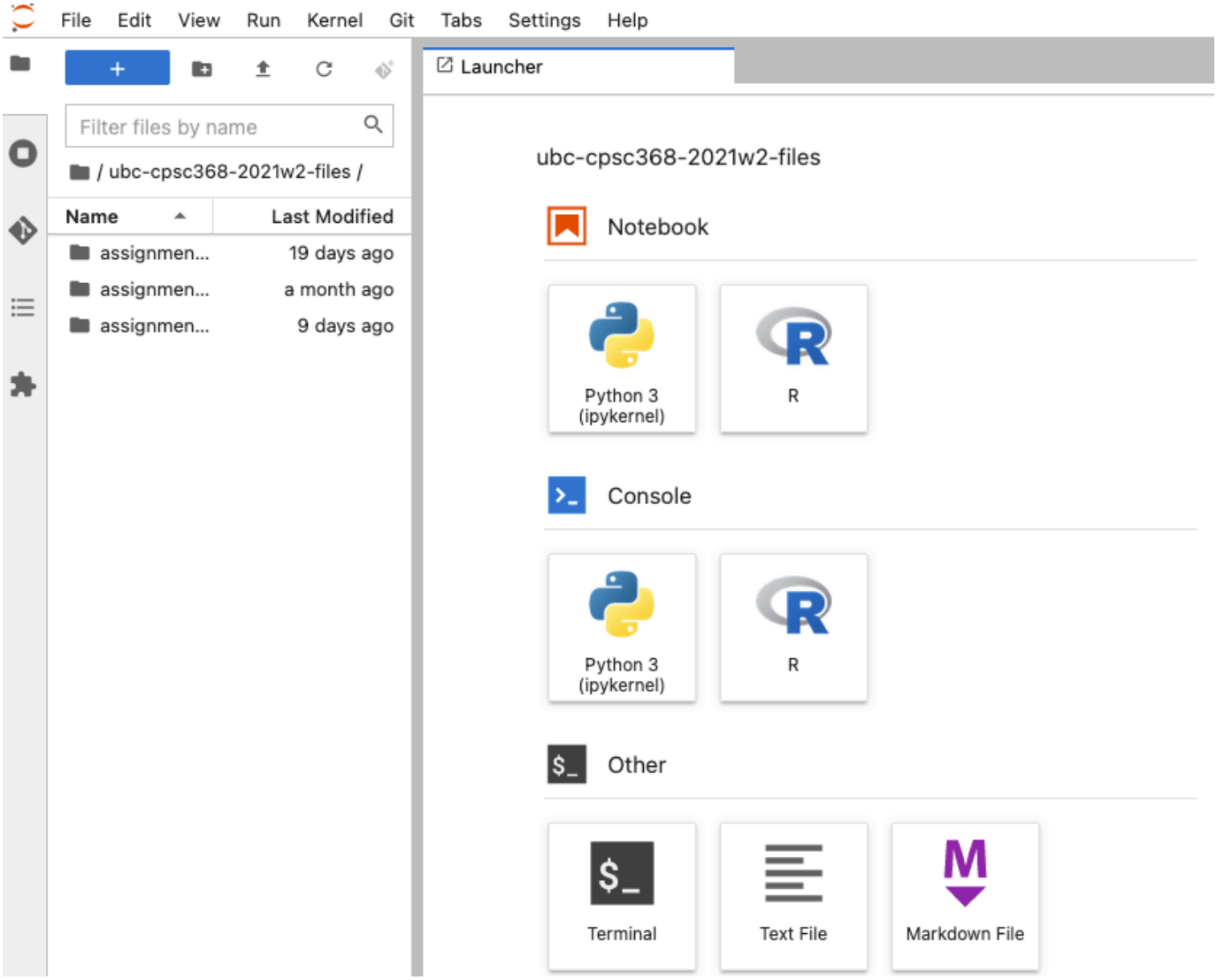
Depending on whether you are working from Syzygy or your own local computer, step 1 will vary. Step 2 will remain the same.

STEP 1: TUNNELLING TO THE DEPARTMENT SERVER

Option 1: If you are working from Syzygy...

1. Go to the launcher view of Jupyter. This is the view with the sidebar that lists all the files in your account (see screenshot below). If you do not see the launcher view of Jupyter,

click on the Jupyter logo in the top left corner of the screen.



2. Click on Terminal.
3. In the resulting Terminal window, type the following and press enter. Be sure to replace `CWL` with your actual CWL id.

   ```
   ssh -l CWL -L 127.0.0.1:1522:dbhost.students.cs.ubc.ca:1522 remote.students.cs.ubc.ca
   ```
4. Enter your CWL password when prompted.

5. After clicking enter, you may be prompted with a message that looks like the following:

   The authenticity of host 'remote.students.cs.ubc.ca (<some IP address>)' can't be established.
   ED25519 key fingerprint is <some series of numbers and letters>.
   Are you sure you want to continue connecting (yes/no/[fingerprint])?

   Type yes and press enter.

6. You must keep this terminal window open while you access the database. When you are finished or want to stop working, type `exit` into the terminal window and hit enter to close the session.
7. Move to step 2 to test that your connection works.

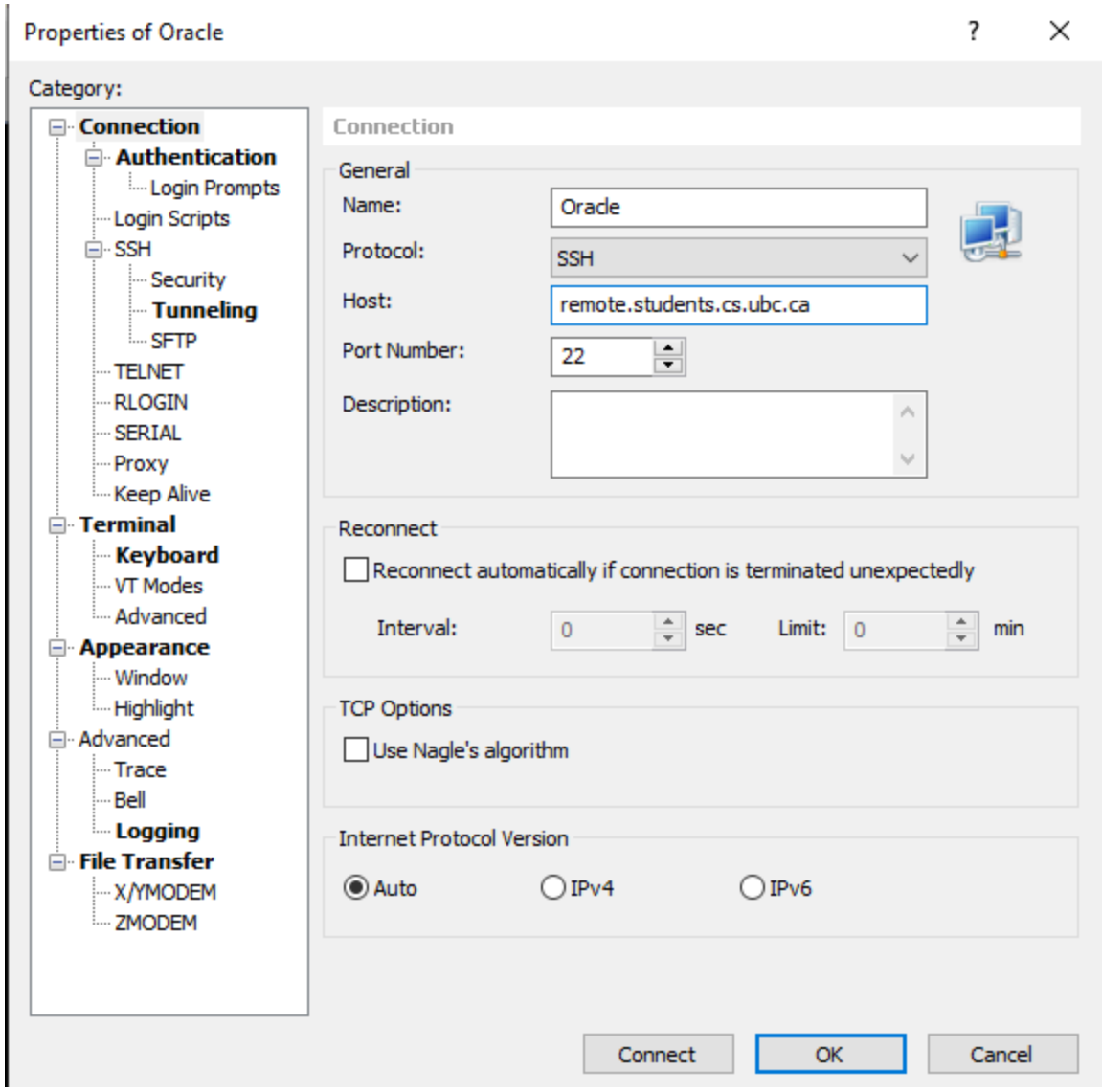
Option 2: If you are working from a local Jupyter Installation...

This section is for anyone who has installed a version on Jupyter on their computer and wants to use this local installation to access the UBC CS department's Oracle database. **It assumes that you have already installed Jupyter, cx_Oracle, and the Oracle Instant Client.** If you have not done these things, please refer to the "Programming Environment" page on Canvas for instructions on how to do so.

WINDOWS

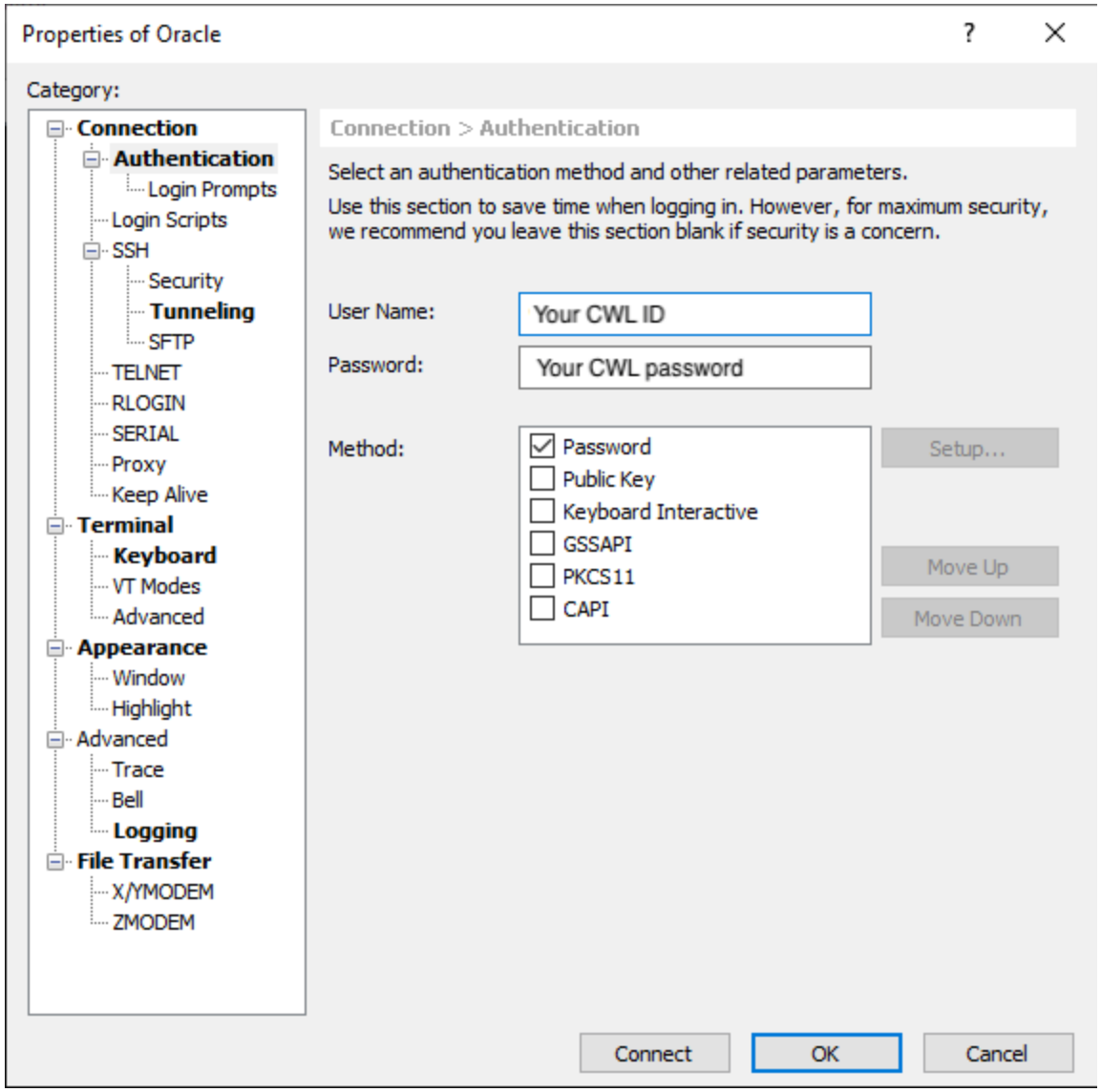
1. If you do not have XManager installed on your computer, download and install a copy.
2. Create an XShell session in XManager by clicking on the leftmost button (the one with a + sign).

3. In **Connection** set the values as per the screenshot below.

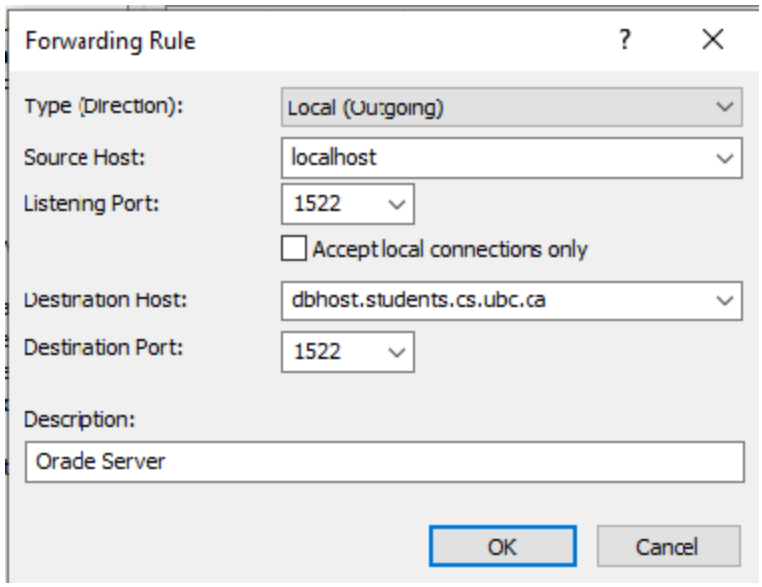


4. In **Authentication**, set the values as per the screenshot below (you can choose to use SSH authentication instead of password but in terms of getting things set up and running

quickly, password might be a better choice). **Use your own CWL username and password.**



5. In **Tunneling**, click add and fill in the information as per the screenshot below.



6. Check the "Forward X11 connection" box. Forward the connection to Xmanager.
   Screenshot of the overall Tunneling screen in XShell.
7. Click "Ok".

8. Launch your connection by clicking on "Oracle" in the sidebar in XManager.
9. Move to step 2 to test that your connection works.

MAC OR UNIX

1. Open Terminal and run:

   ```
   ssh -l username -L localhost:1522:dbhost.students.cs.ubc.ca:1522
   remote.students.cs.ubc.ca
   ```

   Be sure to replace username with your actual CWL username.

2. Move to step 2 to test that your connection works.

STEP 2: TEST YOUR CONNECTION

Assuming you have not removed the tables created by the `bookbiz.sql` file from an earlier assignment, you should be able to test your connection by making a query on one of the bookbiz tables.

Installing the oracledb library
This is something that only needs to be done once. If you have already installed the library, then you can skip this section and move onto the "Connecting to the database" section below.

Install the oracledb liibrary by copying and pasting the following lines into a cell in a Jupyter file and then running it. After installation, you may need to restart the kernel (Kernel -> Restart Kernel).

```
try:
    %pip install --user "oracledb" --no-warn-script-location
except Exception as e:
    print("\x1b[31m\u2717 Unexpected error! Please contact course staff\n" +
         "Please include the entire text above and below in your message.")
    raise
```

Connecting to the database
Try running the following in a Jupyter cell. Be sure to replace `username` with ora_CWLID (no @stu required) and `password` with "a" followed by your student number (e.g., a12345678).

```
import oracledb
dsn = oracledb.makedsn("localhost", 1522, service_name="stu")
connection = oracledb.connect(user="username", password="pwd", dsn=dsn)
```

```
cur = connection.cursor()
for row in cur.execute("select * from titles"):
    print(row)
cur.close()
connection.close()
```